

CENTRALIZED FILE SEARCH SYSTEM

SYSTEMS ANALYSIS REPORT

Executive Summary

The organization's current file-storage environment is decentralized and difficult to navigate using the current search system. Files are distributed across multiple locations without a single, user-friendly search system capable of locating information across the organization's existing repositories. As a result, successful file retrieval depends heavily on employees knowing where specific documents are stored, creating heavy reliance on institutional or tribal knowledge.

This project proposes the implementation of a centralized search system that indexes and retrieves authorized files across multiple existing storage locations without requiring full repository consolidation. The proposed system shall provide users with a simple, intuitive interface for locating documents regardless of their storage location, while enforcing appropriate access controls. This project will support improvements to metadata, naming conventions, and indexing to increase search accuracy and consistency.

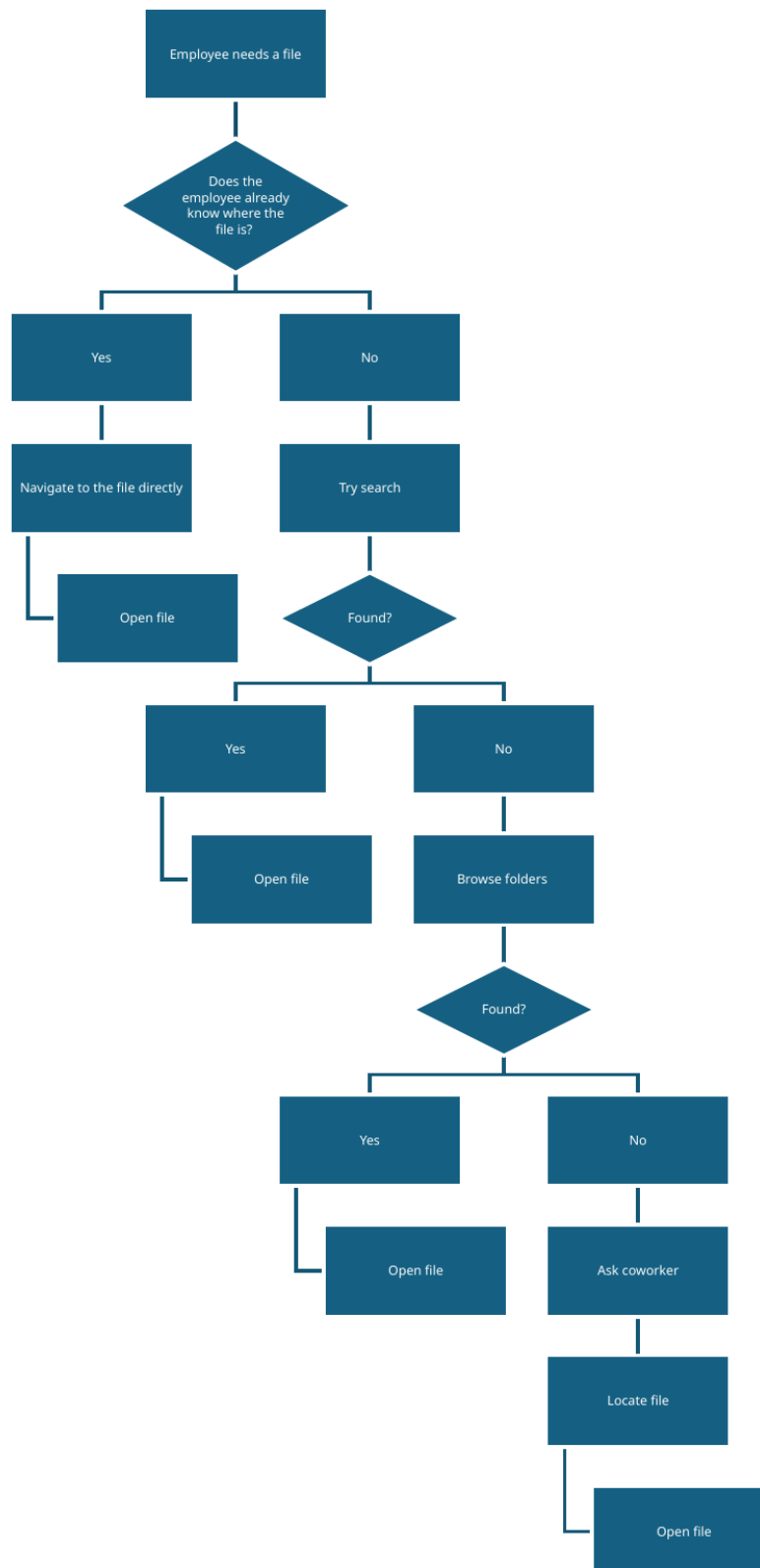
The result will be faster and more reliable information retrieval, greatly reduced dependence on tribal knowledge, improved usability of the organization's existing file-storage environment, and less disruption than would be necessary for a full repository migration and integrated search system.

Current-State Analysis

Overview

The current file system consists of heavily nested folders and files with naming conventions that are unintuitive. This leaves employees who do not already know the location of necessary files unable to locate them without being informed by people who already know. This also means that using the search function is ineffective, as employees cannot intuitively search for the files they need.

AS-IS Analysis



The current file-retrieval process depends heavily on employees already knowing where documents are stored. When the location is unknown, employees must attempt to use the search function, manually browse nested folders, or ask experienced coworkers for assistance. As a result, file retrieval is inconsistent, time-consuming, and dependent on tribal knowledge rather than an intuitive and reliable search process.

Risk Analysis

The current file-retrieval process creates several risks:

- **Loss of institutional knowledge:** Knowledge of where critical files are stored is concentrated among a small number of employees. If those employees leave the organization, the ability to locate certain files may be lost or significantly impaired.
- **Increased file-retrieval time:** When employees who know where files are located are unavailable, other employees may spend substantially more time searching for the information they need.
- **Interruption of primary duties:** Employees with tribal knowledge of the file system may be repeatedly interrupted to help others locate documents, reducing the time available for their primary responsibilities.

Requirements

The proposed system must address the searchability problems of the current file environment and reduce the operational risks created by dependence on employee knowledge.

Business Requirements

- Users must be able to locate necessary files without prior knowledge of their storage location.
- The organization must reduce dependence on individual employees' knowledge of the existing folder structure.
- File retrieval should become faster, more consistent, and easier for both experienced and inexperienced users.
- The proposed solution should improve access to information without requiring unnecessary disruption to existing file repositories.
- The organization must preserve appropriate access restrictions while improving file searchability.

Functional Requirements

- The system shall provide a single search interface capable of searching across multiple authorized file repositories.
- The search interface itself shall be easily locatable.
- The system shall index file names, file contents, and relevant metadata.
- The system shall allow users to search using ordinary keywords rather than requiring exact file names or folder locations.
- The system shall return only files that the user is authorized to access.
- The system shall allow users to filter search results using relevant metadata where available.
- The system shall update the search index when files are added, modified, moved, or removed.
- The search interface shall be simple enough for users to operate without knowledge of the underlying folder structure.
- The system shall support the addition of new repositories without requiring a complete redesign.

Proposed System Design

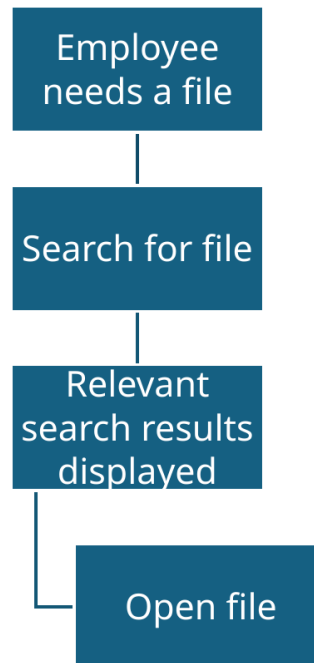
The proposed solution is a centralized search system capable of locating authorized files across multiple existing repositories without requiring the organization to consolidate all files into a single storage location.

An indexing service shall connect to each approved repository and scan file names, file contents, and relevant metadata. This information shall be used to create and maintain a centralized search index. After the initial indexing process is completed, the index shall be updated as files are added, modified, moved, or removed.

Users shall interact with the system through a single, simplified search interface. Instead of navigating through multiple folder structures, users shall enter ordinary search terms related to the information they need. The search system shall query the centralized index and return relevant results from across the connected repositories while preserving existing access permissions.

Where necessary, file metadata and naming conventions shall be standardized or supplemented to improve search accuracy. Relevant metadata may include document type, file name, department, owner, revision, status, effective date, tags, and descriptive keywords.

TO-BE Process



Proposed Architecture

The proposed system shall consist of four primary components:

- **Existing File Repositories:** The current storage locations containing organizational files.
- **Indexing Service:** Scans connected repositories and extracts searchable file content and metadata.
- **Central Search Index:** Stores the searchable representation of files across all connected repositories.
- **Search Interface:** Provides users with a single location from which to search for and access authorized files.

The proposed design allows the organization to improve file discoverability without requiring a disruptive full migration of existing repositories.

Validation and Recommendation

Validation

The proposed system should be validated by confirming that users can locate required files more quickly and with less dependence on tribal knowledge of the existing folder structure.

Validation should include testing whether:

- Users can locate known files using ordinary search terms.
- Search results return the correct and relevant documents.
- Users can locate files without knowing their original storage location.
- Existing access permissions are preserved.
- Newly added or modified files appear in search results after indexing updates.
- Users with limited familiarity with the current file structure can successfully retrieve files using the search interface.
- The average time required to locate files is reduced compared with the current process.

A pilot implementation with a limited number of repositories and users should be conducted before broader deployment. User feedback and search-performance results should be reviewed to identify necessary adjustments to metadata, indexing rules, filters, and search behavior.

Recommendation

The organization should implement a centralized search capability across its existing file repositories rather than immediately consolidating all files into a single storage location. This approach addresses the primary usability problem while reducing the disruption, cost, and migration risk associated with full repository consolidation.

Implementation should begin with an assessment of existing repositories, access controls, naming conventions, and metadata quality. A limited pilot should then be used to validate the proposed architecture and search experience before expanding the system to additional repositories.

If internal staff do not possess the expertise or capacity required to configure and maintain the proposed search system, the organization should consider using a qualified third-party implementation partner.